Spatio-Temporal Graph Neural Network for Motor Movement Classification Using 64-Channel EEG Signals

A General Academic Literature Review and Proposed Methodology

---

Asfar Hossain SItab
ID-2026-2-74-008

Parmita Hossain Simia
ID-2026-2-74-009

Supervisor: Raihan Ul Islam
Associate Professor, Dept. of CSE
East West University (EWU)

July 2026

---

1. Introduction

Electroencephalography (EEG)-based brain–computer interfaces (BCIs) attempt to translate neural activity into commands for rehabilitation, assistive control, and human–computer interaction. Motor imagery (MI) and motor execution are especially important because both produce task-related changes in sensorimotor activity, while MI can be used even when overt limb movement is limited. The classification problem remains challenging because EEG is noisy, non-stationary, subject-dependent, and recorded through multiple spatially distributed electrodes. A successful model must therefore represent not only temporal and spectral patterns within each channel but also relationships among channels.

Convolutional and recurrent networks have improved end-to-end EEG decoding, but their representation of electrode relationships is usually implicit. Graph neural networks (GNNs) provide a more direct formulation: electrodes or derived covariance structures can be treated as nodes, while anatomical proximity, correlation, coherence, phase synchronization, or learned attention can define edges. Message passing then enables the model to combine information from related nodes instead of assuming that EEG channels form a regular image grid.

This report reviews five peer-reviewed papers published from 2022 to 2026 that are closely related to GNN-based motor EEG classification. The selected studies cover five complementary graph-learning directions: feature-level graph embedding, coherence-based graph convolution, manifold-valued graph learning, a forward–forward graph convolutional network, and a dynamic graph-attention framework. The purpose is not to rank papers using raw accuracy alone, because the datasets, task definitions, validation procedures, and participant populations differ. Instead, the review identifies each paper's dataset, application, GNN type, evaluation metrics, strengths, limitations, unresolved research gap, and relationship to the proposed 64-channel spatio-temporal GNN study.

The proposed work will use the PhysioNet EEG Motor Movement/Imagery Dataset and represent its 64 electrodes as graph nodes. Its intended contribution is a leakage-aware spatio-temporal architecture that combines temporal EEG learning with node-level spectral features and a hybrid graph constructed from anatomical proximity and training-only functional relationships. The five reviewed studies provide the closest methodological foundation for this design.

2. Related Work: Five Recent GNN Studies

2.1. Comparative review tables

The following five tables summarize key GNN-based motor EEG studies published between 2022 and 2026. Reported performance values are kept in the context of each paper's own evaluation protocol and should not be interpreted as directly comparable across tables.

Paper 1: EEG_GENet

Table 1: EEG_GENet: feature-level graph embedding for motor imagery.

Aspect	Details
Authors	H. Wang, H. Yu, and H. Wang
Year	2022

Continued on next page

Table 1 – continued from previous page

Aspect	Details
Dataset	BCI Competition IV-2a and High Gamma Dataset
Application	General MI-BCI decoding
Method/model	Graph embedding inserted into EEGNet and other backbones; graphs formed from electrode position or Pearson correlation
Type of GNN	Static feature-level graph embedding / graph convolution component
Metrics	Classification accuracy and improvement over host networks
Strengths	Plug-in design; compares anatomical and correlation graphs; publicly reported implementation
Limitations	Static topology; graph quality depends on hand-designed position or correlation; limited evidence for unseen-subject generalization
Research gap	Dynamic, leakage-controlled, and temporally integrated graph construction remains insufficiently studied
Relation to proposed work	Directly motivates testing anatomical, functional, and hybrid graphs, but the proposed model uses explicit 64-node spatio-temporal message passing

Paper 2: Coherence-Based Graph Convolution Network (C-GCN)

Table 2: C-GCN: coherence-based graph learning for spinal cord injury.

Aspect	Details
Authors	H. Li, M. Liu, X. Yu, J. Zhu, C. Wang, X. Chen, C. Feng, J. Leng, Y. Zhang, and F. Xu
Year	2023
Dataset	Private 64-lead EEG: 18 spinal-cord-injury patients and 7 healthy controls
Application	MI decoding for neurorehabilitation
Method/model	Coherence matrices in μ, β, and combined bands define functional graphs; graph convolution performs classification
Type of GNN	Coherence-based spectral GCN (C-GCN)
Metrics	Accuracy, F1-score, loss; best reported accuracy 96.85% and F1 96.78%
Strengths	Clinically relevant population; frequency-specific functional connectivity; compares several classical and deep baselines
Limitations	Small private dataset; limited reproducibility; no external validation; very high results require cautious interpretation of splitting
Research gap	Need public-data replication, stricter subject-level validation, and comparison with anatomical or hybrid graphs
Relation to proposed work	Supports frequency-aware functional edges for 64-channel EEG; the proposed work adds public data, training-only graph estimation, and subject-independent testing

Paper 3: Graph-CSPNet

Table 3: Graph-CSPNet: GNNs on SPD manifolds.

Aspect	Details
Authors	C. Ju and C. Guan
Year	2024
Dataset	Five public MI-EEG datasets evaluated in 11 subject-specific scenarios
Application	Geometric MI-EEG classification
Method/model	Graph-CSPNet builds a time–frequency graph whose nodes are spatial covariance matrices on symmetric positive-definite manifolds
Type of GNN	Manifold-valued spectral GNN with Graph-BiMap and Riemannian layers
Metrics	Classification accuracy under 10-fold cross-validation and holdout evaluation; near-optimal results in 9 of 11 scenarios
Strengths	Preserves covariance geometry; flexible time–frequency segmentation; strong mathematical interpretability

Continued on next page

Table 3 – continued from previous page

Aspect	Details
Limitations	Complex preprocessing and Riemannian optimization; mainly subject-specific evaluation; nodes are covariance segments rather than electrodes
Research gap	Need simpler electrode-level models, cross-subject evidence, and direct temporal learning from raw multichannel EEG
Relation to proposed work	Provides a geometric alternative and motivates covariance-aware ablations; the proposed work instead models the 64 electrodes directly and fuses raw temporal and spectral node features

Paper 4: F-FGCN

Table 4: F-FGCN: forward–forward graph convolution on PhysioNet.

Aspect	Details
Authors	Q. Xue, Y. Song, H. Wu, Y. Cheng, and H. Pan
Year	2024
Dataset	PhysioNet EEGMMIDB: 109 participants, 64 electrodes, four MI classes; additional BCI Competition III evaluation
Application	Four-class MI decoding
Method/model	Pearson-correlation graph, Chebyshev graph convolution, max pooling, GCN pretraining, and forward–forward dense classification
Type of GNN	Chebyshev spectral GCN with forward–forward learning (F-FGCN)
Metrics	Accuracy; maximum 96.11% subject-level and 82.37% group-level; average 89.39% and 72.81%
Strengths	Uses the same 64-channel public dataset as the proposed study; combines topology with an alternative training mechanism; extensive model comparisons
Limitations	Random 7:2:1 splitting may mix subject information; graph is correlation-based; training is sensitive to negative samples and electrode layouts
Research gap	Strict leave-subject-out validation, training-only graph construction, and temporal-graph ablations are still required
Relation to proposed work	Closest dataset-level comparator; the proposed work adds hybrid anatomical-functional adjacency, explicit temporal blocks, and leakage-aware grouped evaluation

Paper 5: DSSICNN

Table 5: DSSICNN: dynamic GATv2 with spectral and temporal attention.

Aspect	Details
Authors	Z. Shao, Z. Gu, L. Che, Z. Yu, and Y. Li
Year	2026
Dataset	BCI Competition IV-2a and OpenBMI
Application	Session-dependent, session-independent, subject-independent, and online MI decoding
Method/model	DSSICNN combines multi-band processing, temporal log-variance, a PLV graph, dynamic GATv2 spatial guidance, cross-spectral interaction, and temporal attention
Type of GNN	Dynamic graph attention network (GATv2) on a thresholded PLV graph
Metrics	Accuracy, F1-score, Wilcoxon tests, efficiency metrics; e.g., 80.09%/79.79% on IV-2a session-independent and 77.88%/77.56% on OpenBMI
Strengths	Integrates network-neuroscience priors, spectral interaction, temporal attention, ablation, and efficiency analysis without data augmentation
Limitations	Performance depends on PLV threshold; no explicit anatomical prior; graph partitions and graph augmentation remain unexplored
Research gap	Need robust hybrid topology, graph-density sensitivity analysis, and validation on 64-channel motor execution/imagery data

Continued on next page

Table 5 – continued from previous page

Aspect	Details
Relation to proposed work	Motivates an attention-based GNN branch and graph-threshold ablations; the proposed work uses anatomy plus training-only functional coupling and tests GCN, GAT, and GraphSAGE variants

2.2. EEG_GENet: feature-level graph embedding

Wang, Yu, and Wang introduced EEG_GENet as a graph-aware extension of EEGNet rather than designing a completely separate classifier [1]. Their central argument is that conventional convolution processes channel features but does not explicitly represent edges among electrodes. The authors therefore insert a feature-level graph embedding component into EEGNet and also test the component with other network backbones. Two graph definitions are considered: one based on electrode positions and another based on Pearson correlation. This makes the paper important for Track 1 because it distinguishes the GNN component from the surrounding convolutional architecture and demonstrates that graph learning can be treated as a reusable module.

The study is evaluated on the four-class BCI Competition IV-2a dataset and the High Gamma Dataset. Its main metric is classification accuracy, with the graph-enhanced models reported to improve the corresponding host networks. A major strength is the direct comparison of spatial-prior and data-driven graphs. This provides evidence that the construction of the adjacency matrix is not a minor preprocessing decision but a central modeling choice.

The principal limitation is that both graph forms are essentially static. A position graph cannot adapt to task- or subject-specific coupling, while a Pearson graph may be noisy and can cause information leakage when calculated before the training/test split. The paper also does not fully resolve whether graph embedding improves performance for completely unseen subjects. For the proposed work, EEG_GENet motivates three controlled graph conditions—anatomical-only, functional-only, and hybrid—but the functional component will be estimated from training data within each fold. The proposed architecture will also place graph propagation inside temporal blocks so that channel relationships interact with evolving EEG features instead of being used only as a feature-level add-on.

2.3. C-GCN: coherence-based graph learning for spinal cord injury

Li et al. proposed a coherence-based graph convolutional network (C-GCN) for left- and right-hand motor imagery in spinal-cord-injury rehabilitation [2]. The dataset was collected with a 64-lead EEG system from 25 participants, including 18 patients with spinal cord injury and seven healthy controls. Unlike a GCN based only on physical electrode positions, C-GCN uses EEG coherence to define frequency-specific functional relationships among channels. Separate analyses are reported for the μ rhythm, the β rhythm, and their combination.

The model treats coherence-derived multichannel EEG features as graph signals and then uses graph convolution for classification. The paper reports a highest accuracy of 96.85% and an F1-score of 96.78% for the patient data. Average patient-level accuracy was reported as 95.47% in the μ band, 96.14% in the β band, and 97.12% when the two bands were combined. These results support the idea that functional graph structure and motor-related frequency bands contain complementary information. The clinical comparison between spinal-cord-injury participants and healthy controls is another strength because it connects classification with rehabilitation-oriented network analysis.

Nevertheless, the dataset is private and relatively small, which limits independent replication. The very high reported performance should also be interpreted carefully because trial-level cross-validation can be optimistic when participant identity is not isolated across folds. The work does not directly compare coherence graphs with anatomical, correlation, identity, or randomized graph controls under a strict subject-independent protocol. The proposed study addresses these issues using a public 64-channel dataset, training-only graph construction, saved participant-level splits, and explicit graph ablations. C-GCN particularly motivates including motor-band features and investigating whether functional edges align with known sensorimotor regions.

2.4. Graph-CSPNet: GNNs on SPD manifolds

Ju and Guan developed Graph-CSPNet to model motor imagery through geometric deep learning on symmetric positive-definite (SPD) manifolds [3]. Instead of using an electrode as each graph node, the method divides EEG into time–frequency regions, calculates a spatial covariance matrix for each region, and treats each covariance matrix as a manifold-valued node. Edges represent similarity between neighboring time–frequency nodes, measured using Riemannian distance. Graph-BiMap layers, Riemannian batch normalization, logarithmic mapping, and Riemannian optimization preserve the geometry of the covariance representations.

The paper evaluates five public MI-EEG datasets using 10-fold cross-validation and holdout protocols, forming eleven subject-specific experimental scenarios. It reports near-optimal classification accuracy in nine of the eleven scenarios. Its strongest contribution is conceptual: it shows that a graph can represent the time–frequency organization of covariance matrices rather than only physical electrode connectivity. This formulation is especially appropriate when discriminative information lies in spatial covariance patterns and localized ERD/ERS fluctuations.

The method is mathematically sophisticated but computationally and implementation-wise more complex than an ordinary electrode-node GNN. Its evaluation is mainly subject-specific, so it does not fully answer how well the manifold representation transfers to unseen participants. The graph nodes also do not correspond directly to the 64 electrodes, which reduces direct interpretability of channel-level edges. In relation to the proposed work, Graph-CSPNet provides an important geometric baseline and motivates the use of covariance or band-power information in the node-feature branch. The proposed model, however, keeps electrodes as nodes, uses explicit temporal convolution for raw signals, and can therefore visualize learned importance directly on the 64-channel montage.

2.5. F-FGCN: forward–forward graph convolution on PhysioNet

Xue et al. proposed F-FGCN, combining a Pearson-correlation graph, Chebyshev graph convolution, pooling, and a forward–forward learning mechanism [4]. The study is highly relevant because it uses the PhysioNet EEG Motor Movement/Imagery Database, which contains recordings from 109 participants using 64 electrodes. Four motor imagery classes are considered: left fist, right fist, both fists, and both feet. The authors first pretrain a multi-layer GCN and then use forward–forward dense layers for higher-level feature extraction and classification.

The paper reports maximum subject-level and group-level accuracies of 96.11% and 82.37%, respectively. Its table also reports average accuracies of 89.39% at the subject level and 72.81% at the group level. These distinctions are important because maximum and average results answer different questions. The work demonstrates that a spectral Chebyshev GCN can process the same 64-channel graph structure targeted in the proposed research, while the forward–forward mechanism offers an alternative to conventional end-to-end backpropagation.

Two limitations are especially relevant. First, the paper randomly shuffles data and applies a 7:2:1 train/validation/test split, so participant information may be shared between splits. This makes the group result different from a strict leave-one-subject-out experiment. Second, Pearson correlation is used to define the graph, but a correlation matrix estimated using all available trials can leak test-distribution information unless it is rebuilt inside each training fold. The authors also note sensitivity to electrode placement and increased complexity from processing positive and negative samples. The proposed study will use the same dataset but will separate subject-dependent and subject-independent protocols, construct functional edges from training data only, and compare hybrid adjacency against identity, random, fully connected, anatomical-only, and functional-only graphs.

2.6. DSSICNN: dynamic GATv2 with spectral and temporal attention

Shao et al. introduced the Dynamic Spectral-Spatial Interaction Convolution Neural Network (DSSICNN), the most recent paper in this review [5]. The model is evaluated on BCI Competition IV-2a and OpenBMI. It combines a temporal-spectral-spatial convolution branch with a spatial-guidance branch. Phase-locking value (PLV) is used to construct a sparse functional graph, and GATv2 dynamically adjusts the attention assigned to neighboring nodes. Additional modules model cross-spectral interaction and the changing importance of temporal windows during an MI trial.

The paper evaluates session-dependent, session-independent, subject-independent, and online conditions using accuracy, F1-score, statistical testing, parameter count, memory, FLOPs, and runtime. For BCI Competition IV-2a, DSSICNN reports session-independent accuracy and F1-score of 80.09% and 79.79%, and session-dependent values of 86.71% and 86.19%. For OpenBMI, the corresponding session-independent values are 77.88% and 77.56%, while the session-dependent values are 89.00% and 88.72%. A subject-independent BCI Competition IV-2a experiment reports 62.79% accuracy and 62.55% F1-score. The paper also performs component ablation and Wilcoxon signed-rank testing, which strengthens its evaluation.

DSSICNN shows that dynamic attention can compensate for incomplete graph structure and that cross-frequency and temporal interactions are important. However, its initial graph still depends on a manually selected PLV threshold. The authors report that overly sparse and overly dense graphs reduce performance, and they identify graph augmentation and functional subgraph decomposition as future work. For the proposed 64-channel model, this study motivates testing GAT as one GNN type and analyzing graph density. The planned hybrid graph adds a stable anatomical prior to training-only functional coupling, while the dual-branch architecture retains both raw temporal EEG and interpretable spectral node features.

2.7. Overall research gap and positioning of the proposed work

The five papers collectively establish that graph-aware motor EEG models can exploit electrode position, correlation, coherence, covariance geometry, PLV, and learned attention. However, four recurring gaps remain. First, graph construction is often static or calculated without an explicit statement that validation and test trials were excluded. Second, high trial-level accuracy is frequently reported without a strict participant-disjoint protocol. Third, many methods prioritize either raw temporal signals or engineered spectral/covariance features rather than integrating both in a controlled architecture. Fourth, different GNN types and graph definitions are rarely compared under identical preprocessing and data splits.

The proposed work is positioned to address these gaps through a 64-node spatio-temporal graph neural network for the PhysioNet EEGMMIDB. Anatomical nearest-neighbor edges will provide a reproducible structural prior, while functional edges will be computed from training data only and reconstructed for every fold. A temporal branch will learn waveform dynamics, and a node-feature branch will encode motor-related spectral and time-domain descriptors. GCN, GAT, and GraphSAGE variants will be compared, together with anatomical-only, functional-only, hybrid, identity, randomized, and fully connected graphs. Subject-dependent and subject-independent results will be reported separately using accuracy, balanced accuracy, macro-F1, Cohen's kappa, confusion matrices, and participant-level uncertainty.

3. Conclusion

This report reviewed five recent GNN studies published between 2022 and 2026 for motor imagery EEG classification. EEG_GENet demonstrated that graph embedding can improve established EEG networks and highlighted the importance of graph definition. C-GCN showed how frequency-specific coherence can create clinically meaningful functional graphs for spinal-cord-injury rehabilitation. Graph-CSPNet extended graph learning to SPD-manifold covariance representations in the time–frequency domain. F-FGCN provided the closest dataset-level reference by applying a Chebyshev GCN and forward–forward learning to 64-channel PhysioNet EEG. DSSICNN introduced dynamic GATv2-based spatial guidance together with spectral interaction and temporal attention.

The literature supports the use of GNNs but does not establish that one graph type or one reported accuracy is universally superior. Performance depends strongly on the dataset, participant population, task classes, graph construction, preprocessing, and validation design. The clearest unresolved need is a leakage-aware comparison of graph definitions and GNN variants on participant-disjoint data while preserving both raw temporal information and interpretable motor-rhythm features.

The proposed 64-channel ST-GNN will therefore combine anatomical and training-only functional adjacency, temporal graph learning, node-level spectral features, and separate subject-dependent and subject-independent evaluation. This design is directly informed by the strengths of the five reviewed papers while addressing their most important methodological limitations. The resulting study can provide a more defensible assessment of whether hybrid spatio-temporal graph learning improves motor movement classification and generalization in EEG-based BCI research.

References

[1] Wang, H., Yu, H., and Wang, H. (2022). EEG_GENet: A feature-level graph embedding method for motor imagery classification based on EEG signals. *Biocybernetics and Biomedical Engineering*, 42(3), 1023–1040. https://doi.org/10.1016/j.bbe.2022.08.003

[2] Li, H., Liu, M., Yu, X., Zhu, J., Wang, C., Chen, X., Feng, C., Leng, J., Zhang, Y., and Xu, F. (2023). Coherence based graph convolution network for motor imagery-induced EEG after spinal cord injury. *Frontiers in Neuroscience*, 16, 1097660. https://doi.org/10.3389/fnins.2022.1097660

[3] Ju, C., and Guan, C. (2024). Graph neural networks on SPD manifolds for motor imagery classification: A perspective from the time–frequency analysis. *IEEE Transactions on Neural Networks and Learning Systems*, 35(12), 17701–17715. https://doi.org/10.1109/TNNLS.2023.3307470

[4] Xue, Q., Song, Y., Wu, H., Cheng, Y., and Pan, H. (2024). Graph neural network based on brain inspired forward–forward mechanism for motor imagery classification in brain–computer interfaces. *Frontiers in Neuroscience*, 18, 1309594. https://doi.org/10.3389/fnins.2024.1309594

[5] Shao, Z., Gu, Z., Che, L., Yu, Z., and Li, Y. (2026). Dynamic graph based attention spectral network for motor imagery–brain computer interface. *Frontiers in Human Neuroscience*, 20, 1755549. https://doi.org/10.3389/fnhum.2026.1755549